1) Elegir una carta al azar con número k. Tirar k veces el dado. Si sale el 3 gano. Sea X la variable aleatoria que indica el nº de carta

a) Queremos calcular $P(X=2 \mid \text{Franco ganó el juego})$. Aplico Bayes:

$$P(X=2 \mid \text{ganó}) = \frac{P(\text{ganó} \mid X=2) \cdot \underbrace{P(X=2)}_{\frac{1}{10}}}{P(\text{ganó})}$$

$P(\text{ganó} \mid X=2)$ = prob de obtener algún 3 al lanzar el dado dos veces.

$$1 - P(\text{ganó} \mid X=2) = P(\text{no ganó} \mid X=2) = \frac{5}{6} \cdot \frac{5}{6}$$

$$P(\text{ganó} \mid X=2) = 1 - \left(\frac{5}{6}\right)^2 = \frac{11}{36}$$

Ahora calculamos $P(\text{ganó}) \underset{\text{prob total}}{=} \sum_{i=1}^{10} P(\text{ganó} \mid X=i)\, P(X=i)$

$$= \underbrace{\sum_{i=1}^{10} \left(1 - \left(\frac{5}{6}\right)^i\right) \frac{1}{10}}$$

$$P(\text{no ganó} \mid X=i) = \underbrace{\frac{5}{6} \cdot \frac{5}{6} \cdots \frac{5}{6}}_{i\text{-veces}} = \left(\frac{5}{6}\right)^i$$

$$P(\text{gano}) = \sum_{i=1}^{10} \left(\frac{1}{10} - \frac{1}{10}\left(\frac{5}{6}\right)^i\right) = 1 - \frac{1}{10}\sum_{i=1}^{10}\left(\frac{5}{6}\right)^i$$

$$= 1 - \frac{1}{10}\cdot\frac{5}{6}\sum_{i=1}^{10}\left(\frac{5}{6}\right)^{i-1}$$

$$= 1 - \frac{1}{10}\cdot\frac{5}{6}\sum_{j=0}^{9}\left(\frac{5}{6}\right)^{j}$$

$$= 1 - \frac{1}{\cancel{10}_{2}}\cdot\frac{\cancel{5}}{6}\cdot\frac{1-\left(\frac{5}{6}\right)^{10}}{1-\frac{5}{6}}$$

$$= 1 - \frac{1}{2}\left(\frac{1}{6}\,\frac{\left(1-\left(\frac{5}{6}\right)^{10}\right)}{\frac{1}{6}}\right)$$

$$= 1 - \frac{1}{2}\left(\left(1-\left(\frac{5}{6}\right)^{10}\right)\right)$$

$$= 1 - \frac{1}{2}\left(1-\left(\frac{5}{6}\right)^{10}\right)$$

$$= 1 - \frac{1}{2} + \frac{1}{2}\left(\frac{5}{6}\right)^{10}$$

$$= \frac{1}{2}\left(1+\left(\frac{5}{6}\right)^{10}\right)$$

$$P(X=2 \mid \text{gano}) = \frac{P(\text{gano} \mid X=2)\,P(X=2)}{P(\text{gano})}$$

$$P(X=21 \text{ ganó}) = \frac{11}{180\left(1+\left(\frac{5}{6}\right)^{10}\right)}$$

b) Fromo repite el experimento 7 veces. Prob de que haya ganado al menos 3 veces.

Y = Cantidad de veces que ganó.

$$Y \sim \text{Binomial}(7, p) \rightarrow p = P(\text{ganó}) = \frac{1}{2}\left(1+\left(\frac{5}{6}\right)^{10}\right) \approx 0{,}58$$

$$P(Y \geq 3) = 1 - P(Y=0) - P(Y=1) - P(Y=2)$$

$$= \left[1 - \sum_{i=0}^{2} \binom{7}{i} p^i (1-p)^{7-i}\right]\Bigg|_{p=0{,}58}$$

c) 9/5/21 — 13/5/21

Z = nº de repeticiones hasta el primer éxito

$$\underbrace{9, 10, 11, 12}_{\text{perdió}}, \underbrace{13}_{\text{ganó}}$$

$$Z \sim \text{Geométrica}(p)$$

$P(\text{gana por 1}^{\text{er}} \text{ vez el } 13/5/21) = p(1-p)^4$

2) $(a, b, a, b, \ldots \mid, a, a)$

$\#a = \#b$

2a) Prueba de que el juego es finito

- Observación: $X =$ cantidad de partidas ganadas de Ana

$$\#\text{total de partidas} = \begin{cases} X + (X+2) & \text{, si Beto ganó} \\ X + (X-2) & \text{, si Ana ganó} \end{cases}$$

El nº total de partidas es un nº par.

Sea A_{2k} = "Ana ganó en la ronda nº $2k$-ésima"

$(a, b, b, a, \ldots)$

$(a, b, a, b,$

$$P(A_{2k}) = \underbrace{(2p(1-p))}^{k-1} \underbrace{p^2}$$

En los dos primeros: ganó 1 Ana y ganó 1 Beto.

$$= p(1-p) \rightarrow (a,b)$$

$$(1-p)\,p \rightarrow (b,a)$$

$$= 2p(1-p).$$

$$\left(\underbrace{a,b}_{2},\ \underbrace{b,a}_{4}\ \underbrace{\quad}_{6}\ \underbrace{\quad}_{\cdots}\ \underbrace{\quad}_{2(k-1)}\ \underbrace{\quad}_{2k}\right)$$

$A_{2k} \rightarrow$ 2k rondas

B_{2k} = "Beto ganó en la partida n° 2k-ésima"

$$\Rightarrow P(B_{2k}) = (2p(1-p))^{k-1}(1-p)^2$$

$$P(\text{el juego finito}) = P(A_2 \cup B_2 \cup A_4 \cup B_4 \cup \ldots)$$

$$P(\text{el juego finito}) = \sum_{k=1}^{\infty} P(A_{2k}) + P(B_{2k})$$

$$= \sum_{k=1}^{\infty} (2p(1-p))^{k-1} p^2 + (2p(1-p))^{k-1} (1-p)^2$$

$$= \sum_{k=1}^{\infty} (2p(1-p))^{k-1} \left(p^2 + (1-p)^2\right)$$

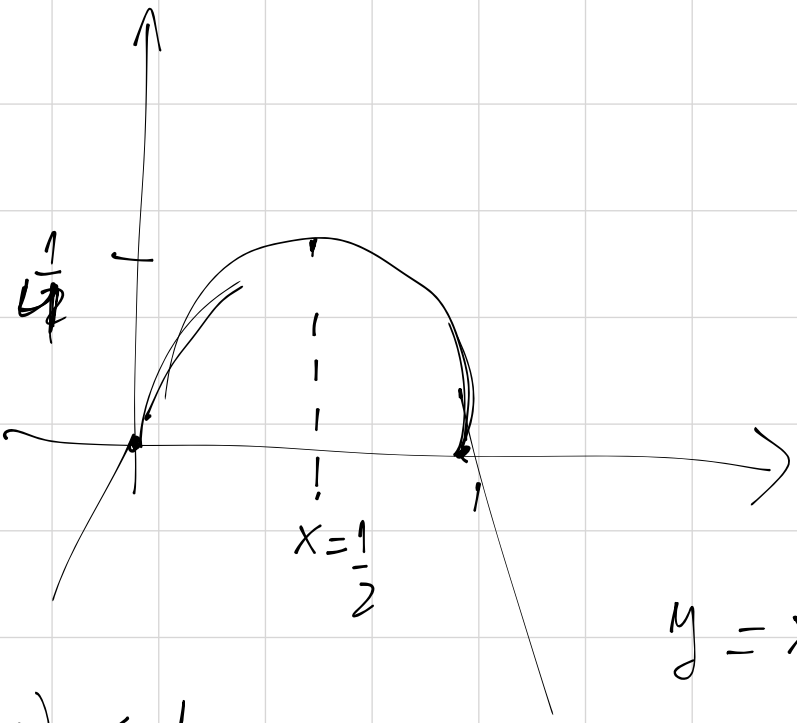
$$= p^2 + (1-p)^2 \sum_{k=1}^{\infty} (2p(1-p))^{k-1}$$

$$2\underbrace{p(1-p)}_{<\frac{1}{2}} < 1$$

$$= \left(p^2 + (1-p)^2\right) \frac{1}{1 - 2p(1-p)}$$

$$= \frac{p^2 + 1 - 2p + p^2}{1 - 2p + 2p^2} = \boxed{1}$$

$2x(1-x)$ $\qquad 0 < x < 1$



$$f\left(\frac{-b}{2a}\right) = f\left(\frac{-1}{-2}\right)$$

$$= f\left(\frac{1}{2}\right)$$

$$\frac{1}{2}\left(1-\frac{1}{2}\right) = \frac{1}{4}$$

$$y = x(1-x)$$

$$= -x^2 + x$$

$$\frac{-b}{2a}$$

$$2x(1-x) \leq \frac{1}{2}$$

b) $P(\text{gano Ana}) = P(A_2 \cup A_4 \cup A_6 \dots)$

$$= \sum_{k=1}^{\infty} P(A_{2k}) = \sum_{k=1}^{\infty} (2p(1-p))^{k-1} p^2$$

$$= \frac{p^2}{1-2p(1-p)} = \frac{p^2}{p^2+(1-p)^2}$$

c) $(\underbrace{a, b, b, a, \dots}_{\geq 8}$ $\qquad (a, a)$

$$P(A_2 \cup B_2) = P(A_2) + P(B_2) = p^2 + (1-p)^2$$

$$P(A_4 \cup B_4) = (2p(1-p))^1 p^2 + (2p(1-p))^1 (1-p)^2 =$$

$$(2p(1-p))\,(p^2+(1-p)^2)$$

$$P(A_6 \cup B_6) = (2p(1-p))^2 p^2 + (2p(1-p))^2 (1-p)^2$$

$$(2p(1-p))^2\,(p^2+(1-p)^2)$$

$$+ \quad = \boxed{(p^2+(1-p)^2)\,(1 + 2p(1-p) + (2p(1-p))^2)}$$

$$P(\text{juego dura más de } 7) = 1 - P(\text{"2"}) - P(\text{"4"}) - P(\text{"6"})$$

$$= 1 - \left[\left(p^2 + (1-p)^2\right)\left(1 + \left(2p(1-p)\right) + \left(2p(1-p)\right)^2\right)\right]$$

$$= 1 - \left[\left(p^2 + (1-p)^2\right)\left(1 + 2p - 2p^2 + 4p^2(1-p)^2\right)\right]$$

$$= 1 - \left[\right.$$

$$= \left(2p(1-p)\right)^3$$

$$P(A_2 \cup A_4 \cup A_6 \ldots)$$

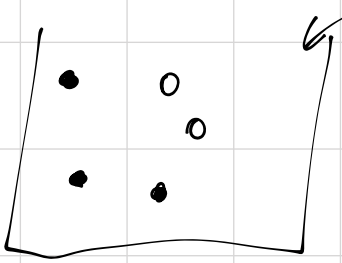
d) $P(\text{gana Ana} \mid \text{el juego durará} + 7) = P(\text{gana A})$?

$$\frac{P(\text{gana A} \cap \text{juego durará} + 7)}{P(\text{juego durará} + 7)} = \frac{P(A_8 \cup A_{10} \cup A_{12} \ldots)}{P(A_8 \cup B_8 \cup A_{10} \cup B_{10} \ldots)}$$

$$(A_2 \cup A_4 \cup A_6 \ldots) \cap (A_8 \cup B_8 \cup A_{10} \cup B_{10} \ldots)$$

$$= A_8 \cup A_{10} \cup A_{12} \cup A_{14}$$

$$\frac{\sum_{k=4}^{\infty} \left(2p(1-p)\right)^{k-1} p^2}{\sum_{k=4}^{\infty} \left(2p(1-p)\right)^{k-1} \cdot \left(p^2+(1-p)^2\right)} = \underbrace{\frac{p^2}{p^2+(1-p)^2}}_{P(\text{gano Ana})}$$

3) 

Bolas 2 (sin repo)

Y = cant. bolitas negras

$$X \sim N(\mu = 12,\ \sigma^2 = 25) \Rightarrow X \perp Y$$

$$P(X^Y \leq 25) = \sum_{j=0}^{2} P(X^Y \leq 25 \mid Y=j)\, P(Y=j)$$

$$= \sum_{j=0}^{2} P(X^j \leq 25 \mid Y=j)\, P(Y=j)$$

$$= \boxed{P(X^0 \leq 25)\, P(Y=0) + P(X \leq 25)\, P(Y=1) + P(X^2 \leq 25)\, P(Y=2)}$$

Hallo Y:

$$P(Y=0) = \frac{2}{5} \cdot \frac{1}{4} = \frac{2}{20} = \frac{1}{10}$$

$$P(Y=1) = \frac{2}{5} \cdot \frac{3}{4} + \frac{3}{5} \cdot \frac{2}{4} = \frac{12}{20} = \frac{3}{5}$$

$$P(Y=2) = \frac{\binom{3}{2}\binom{2}{0}}{\binom{5}{2}} = \frac{3!}{\cancel{2!}1!} \cdot \frac{\cancel{2!}\,3!}{5!} = \frac{3!\,3!}{5!} = \frac{2.3}{4.5} = \frac{6}{20} = \frac{3}{10}$$

Mírnomor X:

$$X \sim N(12, 25)$$

$$P(X^0 \leq 25) = 1$$

$$\frac{X-12}{5} \sim N(0,1)$$

$$P(X \leq 25) = P\left(\frac{X-12}{5} \leq \frac{13}{5}\right) = \varphi\left(\frac{13}{5}\right) \approx 0,995$$

$$P(X^2 \leq 25) = P\left(\sqrt{X^2} \leq \sqrt{25}\right) = P(|X| \leq 5)$$

$$= P(-5 \leq X \leq 5) = P\left(\frac{-5-12}{5} \leq \frac{X-12}{5} \leq \frac{5-12}{5}\right)$$

$$= \left(-\frac{17}{5} \leq (\ ) \leq -\frac{7}{5}\right) = \varphi\left(-\frac{7}{5}\right) - \varphi\left(-\frac{17}{5}\right)$$

$$= \left(1 - \varphi\left(\frac{7}{5}\right)\right) - \left(1 - \varphi\left(\frac{17}{5}\right)\right)$$

$$= \varphi\left(\frac{17}{5}\right) - \varphi\left(\frac{7}{5}\right) \approx 0,08$$

4) Hay 5 personas

T_i = tiempo que tarda la persona i en resolver el problema.

$$T_i \sim iid \ \frac{Y}{X} \quad , \quad Y \sim Exp\left(\frac{1}{2}\right)$$
$$X \sim U[2,4]$$

D = duración del Torneo $\qquad T_1 \sim \frac{Y}{X}$

$$F_D(t) = P(D \leq t) = P\left(\min_{1 \leq i \leq 5} (T_i) \leq t\right) =$$

$$1 - P(T_1 > t, T_2 > t, \dots, T_5 > t) =$$

$$1 - P(T_1 > t) P(T_2 > t) \dots P(T_5 > t) =$$

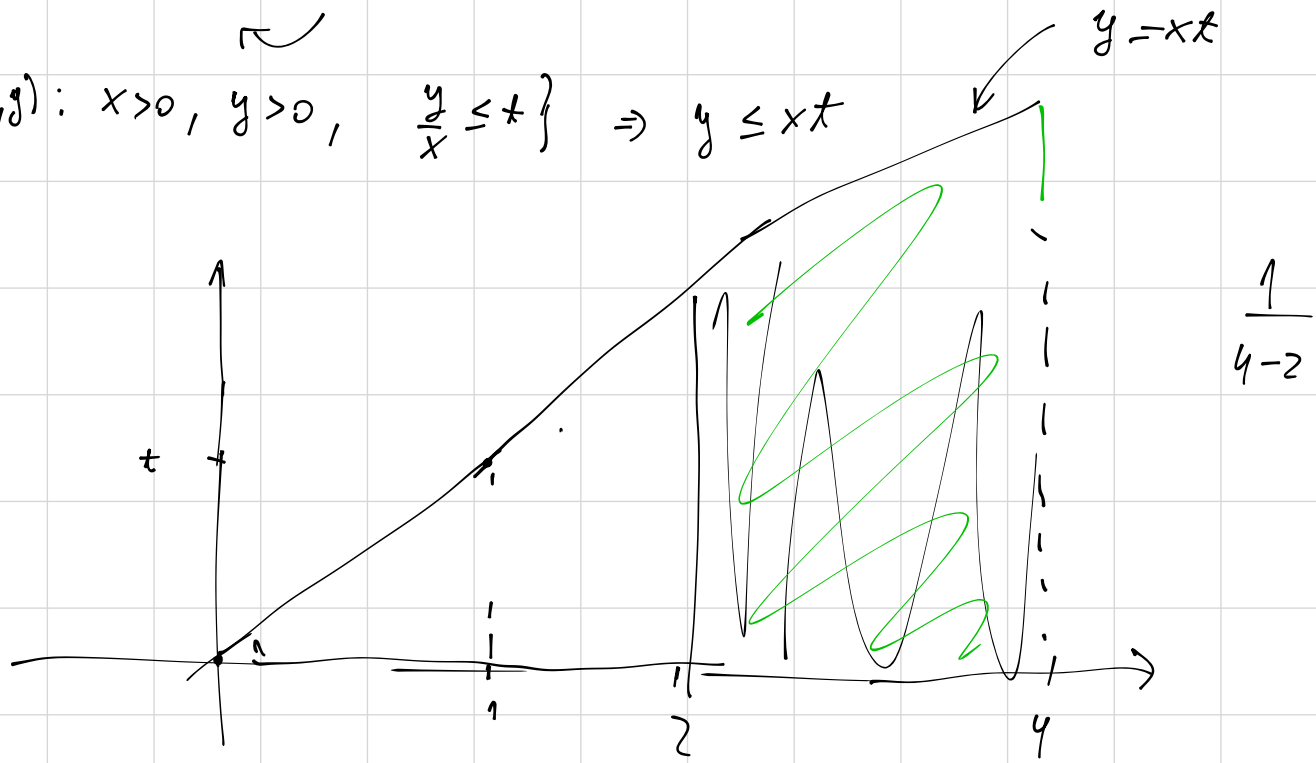
$$1 - (1 - F_{T_1}(t))^5 = 1 - \left(1 - F_{\frac{Y}{X}}(t)\right)^5$$

2° paso) Calcular $F_{\frac{Y}{X}}(t) = \begin{cases} 0 & , t < 0 \\ & , X \sim U[2,4] \\ & , t \geq 0 \quad Y \sim Exp\left(\frac{1}{2}\right) \end{cases}$

Caso $t>0$

$$P\left(\frac{Y}{X}\le t\right) = \int_2^4\int_0^{xt} \frac{1}{2}\,\frac{1}{2}\cdot e^{-\frac{1}{2}y}\,dy\,dx \qquad X\sim U$$

$$\left\{(x,y):\ x>0,\ y>0,\ \frac{y}{x}\le t\right\} \Rightarrow y\le xt$$



$$= \frac{1}{2}\int_2^4\int_0^{xt} \frac{1}{2}e^{-\frac{1}{2}y}\,dy\,dx$$

$$= \frac{1}{2}\int_2^4 F_Y(xt)\,dx = \frac{1}{2}\int_2^4\left(1-e^{-\frac{1}{2}xt}\right)dx$$

$$\left[1+\frac{e^{-2t}}{t}-\frac{e^{-t}}{t}\right] \qquad = \frac{1}{2}\left(x+\frac{2e^{-\frac{1}{2}tx}}{t}\right)\Bigg|_2^4$$

$$= \frac{1}{2}\left(4+\frac{2e^{-2t}}{t}-2-\frac{2e^{-t}}{t}\right)$$

$$F_{\frac{Y}{X}}(t) = 1 + \frac{e^{-2t}}{t} - \frac{e^{-t}}{t}$$

$$F_D(t) = 1 - \left(1 - F_{\frac{Y}{X}}(t)\right)^5$$

$$= 1 - \left(1 - \left[1 + \frac{e^{-2t}}{t} - \frac{e^{-t}}{t}\right]\right)^5$$

$$= 1 - \left(\frac{e^{-t}}{t} - \frac{e^{-2t}}{t}\right)^5$$

$$F_D(t) = \begin{cases} 0, & t < 0 \\ (\ ), & t = 0 \\ 1 - \left(\frac{e^{-t}}{t} - \frac{e^{-2t}}{t}\right), & t > 0 \end{cases}$$

$$\lim_{t\to 0^+} 1-\left(\frac{e^{-t}}{t}-\frac{e^{-2t}}{t}\right)$$

$$\lim_{t\to 0^+} \frac{t-e^{-t}+e^{-2t}}{t} \underset{L'H}{=} \frac{1+\overset{1}{e^{-t}}-2\overset{1}{e^{-t}}}{1}$$

$$= \frac{1+1-2}{1} = 0$$

$$\lim_{t\to+\infty} F_D(t) = \lim_{t\to+\infty} 1-\left[\frac{e^{-t}}{t}-\frac{e^{-2t}}{t}\right]$$

$$= 1-\underbrace{\lim_{t\to+\infty} \frac{e^{-t}-e^{-2t}}{t}} = 1-0 = 1$$

$$= \frac{\underset{0}{-e^{-t}}+2e^{-2t}\to 0}{1}$$

5) $\lambda > 0$, $\quad X, Y \sim iid\ Exp(\lambda)$

a) $U = \dfrac{X}{X+Y}$, $\quad \dfrac{X}{X+Y}$, $X+Y$

$$\downarrow \qquad\qquad \downarrow$$

$$g : (0,+\infty)^2 \longrightarrow (0,1)\times(0,+\infty)$$

$$g\begin{pmatrix} X \\ Y \end{pmatrix} = \begin{pmatrix} \dfrac{X}{X+Y} \\ X+Y \end{pmatrix}$$

$$\frac{u}{u+v} = \frac{w}{w+z} \quad y \quad u+v = w+z$$

$$\frac{u}{w+z} = \frac{w}{w+z} \Rightarrow u = w$$

$$\Rightarrow v = z$$

$$g^{-1}\begin{pmatrix} \dfrac{X}{X+Y} \\ X+Y \end{pmatrix} = \begin{pmatrix} X \\ Y \end{pmatrix}$$

$$w = \frac{X}{X+Y} \Rightarrow wz = X$$

$$z = X+Y \Rightarrow Y = z - wz$$
$$= z(1-w)$$

$$g^{-1}\begin{pmatrix} w \\ z \end{pmatrix} = \begin{pmatrix} wz \\ z(1-w) \end{pmatrix}$$

$$J(g^{-1}) = \begin{array}{c} \begin{array}{cc} \partial w & \partial z \end{array} \\ \begin{vmatrix} z & w \\ -z & 1-w \end{vmatrix} \end{array} = z(1-w) + wz = z$$

$$|J(g^{-1})| = |z| = z$$

$$f_{X,Y}(x,y) = \lambda e^{-\lambda x} \cdot \lambda e^{-\lambda y} \boxed{\mathbb{1}_{(0,+\infty)}(x)\, \mathbb{1}_{(0,+\infty)}(y)}$$

Teorema del Jacobiano:

$x = wz$
$y = z(1-w)$

$$f_{W,Z}(w,z) = \lambda e^{-\lambda wz} \lambda e^{-\lambda z(1-w)} \cdot z\, \mathbb{1}_{(0,+\infty)}(wz)\, \mathbb{1}_{(0,+\infty)}(z(1-w))$$

$$= \lambda^2 \cancel{e^{-\lambda wz}} \cdot e^{-\lambda z} \cdot \cancel{e^{\lambda zw}}\, z$$

$$= \lambda^2 z e^{-\lambda z} \cdot \mathbb{1}_{(0,+\infty)}(wz)\, \mathbb{1}_{(0,+\infty)}(z(1-w))$$

$0 < wz$
$0 < z(1-w)$ $\Rightarrow$ $\boxed{\overset{w}{(0,1)} \times \overset{z}{(0,+\infty)}}$

$$= \lambda^2 z e^{-\lambda z}\, \mathbb{1}_{(0,1)}(w)\, \mathbb{1}_{(0,+\infty)}(z)$$

$g : (0,+\infty)^2 \to (0,1) \times (0,+\infty)$

b) $P(U < X+Y \mid U > \frac{1}{2}) =$ $\qquad U \sim U(0,1)$

$$= \frac{P\left(\{U < X+Y\} \cap \{U > \frac{1}{2}\}\right)}{\underbrace{P(U > \frac{1}{2})}_{= \frac{1}{2}}} = 2\, P\left(\{U < \underset{\substack{\downarrow \\ Z \sim \text{Gamma}(\lambda, 2)}}{X+Y}\} \cap \{U > \tfrac{1}{2}\}\right)$$

$$= 2 \int_{\frac{1}{2}}^{1} \int_{u}^{+\infty} \frac{1}{2} \lambda^2 z e^{-\lambda z} \, dz \, du$$

$$\{(x,y) ; 0 < x < 1, \quad x > \tfrac{1}{2}, \quad y > x\}$$

$$= \frac{2}{2} \int_{\frac{1}{2}}^{1} \left[-\frac{1}{\lambda} \underset{\substack{\downarrow \\ 0}}{e^{-\lambda z}} (\lambda z + 1) \right]_{u}^{+\infty}$$

$$= \int_{\frac{1}{2}}^{1} \frac{1}{\lambda} e^{-\lambda u} (\lambda u + 1) \, du$$

$$= \frac{1}{\lambda} \int_{\frac{1}{2}}^{1} e^{-\lambda u} (\lambda u + 1) \, du$$

$$= \frac{1}{\lambda} \left[e^{-\lambda u} (u+1) \right]_{\frac{1}{2}}^{1} = \frac{1}{\lambda} \left[2 e^{-\lambda} - \frac{3}{2} e^{-\frac{\lambda}{2}} \right]$$

$g : G_0 \longrightarrow G$

$P(g \in G_0) = 1$

$\mathbb{R}^2 \to \qquad f_{x,y}(x,y) = (\ )$

$\boxed{(0,+\infty)\times(0,+\infty)} \qquad f_{x,y}(x,y) = (\ )$

$$f_{w,z}(\ ) \begin{cases} \ , & (w,z) \in G \\ 0 , & (w,z) \in G^c \end{cases}$$

$$2 \quad (a, a) \rightarrow p \cdot p = p^2$$

$$4 \begin{cases} (a, b, a, a) \rightarrow p \cdot (1-p) p p = p(1-p)p^2 \\ (b, a, a, a) \rightarrow (1-p) p \cdot p \cdot p = p(1-p) p^2 \end{cases} \quad + = 2$$

$$6 \begin{cases} (a, b, b, a, a, a) \rightarrow 4 p^2 (1-p)^2 p^2 \\ (a, b, a, b, a, a) \\ (b, a, b, a, a, a) \quad (2 p (1-p))^2 p^2 \\ (b, a, a, b, a, a) \quad 2^2 p^2 (1-p)^2 \end{cases}$$